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Thermally Induced Microcracking in Granites: Characterization and Analysis

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Senones and Remiremont granite samples were thermally cracked at temperatures ranging from 20° to 600°C and subjected to scanning electron microscopy (SEM) observations. Crack densities were quantified using direct length and width measurements. SEM observations showed that during the thermal treatment, the crack lengths barely evolved whereas their width increased with increasing cracking temperature intensity. The correlation between crack porosity, derived from SEM morphological observations and that determined by the compressibility test is most satisfactory. This test is, thus, reliable for the volume characterization of cracks. By contrast, it is impossible to deduce the length and width of the cracks. This study of the mechanical behaviour of thermally cracked granites indicated a negative Poisson ratio in compression and tension. It is most probable that the stresses which remained in the material during cooling caused these abnormalities.

1. INTRODUCTION

The mechanical behaviour of rocks, essentially depends upon the following five factors: mineralogy, structure, temperature, stress and time. Rock mechanics has only recently considered the factor of temperature as important. In many energy resource applications, where rocks are heated or cooled, for geothermal energy recovery and nuclear waste disposal studies, a fundamental understanding of thermal cracking mechanisms is imperative.

Small cracks can influence substantially the physical properties of rocks [1–5]. In the present study, two granites of different granularity were thermally cracked at room pressure. Measurements were taken by using a Scanning Electron Microscope (SEM) and a statistical determination of microcracks was made. Observations carried out (at room temperature) were compared with compressibility tests of the samples after thermal cycling.

Also in this paper results from compressive and tensile mechanical tests performed on cores after thermal cycling are reported. The mechanical parameter values are shown to be dependent upon the density of microcracks in the rocks.

2. EXPERIMENTAL METHODS

2.1. Rock types

Two granites, identical mineralogically, but with different granulometry were selected (Table 1). The Senones granite has a porphyritic texture. Quartz occurs as small, equidimensional crystals at the periphery of feldspars. K-feldspar often occurs as large as 10 mm crystals, whereas plagioclase crystals are smaller. Biotite is found in 2–3 mm crystals and larger than amphibol. The Remiremont is fine grained with a homogeneous texture. Quartz is arranged in groups, the K-feldspar being small and plagioclase being subhedral, prismatic and zoned. Muscovite is lamellar. Complementary details concerning these rocks are examined by Homand-Etienne [6].

2.2. Sample preparation and thermal treatment

Cylindrical cores of granite, 50 mm in diameter and 100 mm long, were taken from the same block of each rock. Homogeneous samples were selected from density determinations and longitudinal wave velocity measurements.

In order to avoid provoking thermal shock, the cores were heated slowly. At a rate of 50°C hr^{-1} the temperature was increased up to 300°C; thereafter, the rate was increased to 100°C hr^{-1}. The maximum temperature was maintained for 5 hr. Different samples were

Table 1. Mineralogical composition (M.C.) and maximum grain-size (G.S.) of Remiremont and Senones granites

	Remiremont granite		Senones granite	
	M.C. (%)	G.S. (mm)	M.C. (%)	G.S. (mm)
Quartz	25–27	2	14–20	2
K-Feldspar	22–24	2	34–45	15
Plagioclase	41–45	1	30–35	2
Biotite	5	0.5	5–10	2
Muscovite	3	0.5	—	—
Amphibole	—	—	2–5	1

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thermally treated to 200°, 400°, 500° and 600°C and then put into a desiccator until they were tested.

For the SEM observations, 20 mm diameter cores were taken from the centre of both untreated and thermally treated cylindrical samples. These cores were then cut at 2 cm below their upper surface for SEM examination. The surfaces were polished with diamond paste. Specimens, were then coated with approximately 100 Å gold–palladium. Finally they were examined by a Stereoscan SEM 250.

3. ELECTRON MICROSCOPY OBSERVATIONS

3.1. Qualitative analysis

The approximately isometric cavities of pores, observed in the crystals have either circular or angular shapes. They occur abundantly in micas and plagioclase and, as Montgomery and Brace [7] observed, less frequently in potassic feldspar. With the two crystals in contact, the intercrystalline boundaries could be entirely sealed off (marked uniquely by the change of contrast). For the most part, they are marked by pores, separated by material bonds. Additionally, a real intercrystalline cracking exists, notably between crystals of different mineralogical nature. Generally, the crack tips are rounded. Intracrystalline cracks are also present. Within quartz crystals, the latter occur in any direction; whereas, in feldspars, they generally follow the cleavage planes. The thermal treatment excites all the initial discontinuities and new cracks are created as a result. From 200°C upwards, the Senones and Remiremont granite intercrystalline boundaries open partially (new intercrystalline cracks). At 600°C, it is possible to estimate that all crystals are surrounded by the interrupted cracks of weak material bonds. In the plagioclase and potassic feldspar, the pores become more numerous. They, however, are formed later in quartz. Generally, the new intracrystalline cracks form in the feldspar from the intruded mismatches present in the crystal such as plane of macles, cleavages and pre-existing discontinuities. These intracrystalline cracks develop progressively but are most pronounced between 500° and 600°C. These two granites demonstrate an identical, discontinuity network evolution.

3.2. Quantitative analysis

Cracks were studied by microphotographic analysis that enlarged the Senones granite by 60 and that of Remiremont by 120. The observed areas (from 29 to 2 mm^2) were determined in relation to the grain-size and the apparent cracking density. (Initially, measurements were taken on non-treated granites; thereafter, at 200°, 400°, 500° and 600°C). As far as possible, similar zones were selected from different samples to prepare the sections. Accordingly, quantitative petrostructural analysis methods were adapted based upon those, proposed by Warren [8]. Such analytical methods deal with the acquisition of topographical data from the digitalization of photographs. In order to obtain and process the crack data, a series of Apple II microcomputer programs were used.

Each crack tip was digitized, enabling the coordinates, the length and orientation to be calculated. Hadley's [9] definition of the crack was used. This considers the crack as a mismatch of width/length ratio, inferior to 0.1, whose orientation does not alter more than 20°C on any significant section of its length. The crack lengths measured vary between 14 and 1 100 μm. Obviously, smaller cracks, invisible at the study scale used for this study, exist, as well as larger ones, in particular those occurring in the extremely large Senones granite feldspars. After a detailed calibration of numerous direct SEM measurements, each crack width was measured on the photograph. The parameter of shape is calculated by entering these data for each crack. The error in crack width measurement is significant (3–4 orders of magnitude). This measurement error is enlarged by the uncertainty concerning real width definition. In fact, relative to their dip, cracks obliquely cutting the measurement plane, make a real correction impossible. Correction, thus, by obliquity is purely random. The real C width of an open discontinuity C^* is $C = C^* \sin\theta$, θ specifying the discontinuity dip angle with respect to the measurement plane. Assuming that the distribution of cracks is random, there is a uniform probability in space for a θ to be included between 0 and $\pi/2$. Therefore, the average value of $\sin\theta$ is:

$$\overline{\sin\theta} = \frac{\int_0^{\pi/2} \sin\theta \, \mathrm{d}\theta}{\pi/2} = 2/\pi.$$

From the above it follows that the corrected expression for average crack widths is:

$$C = 2C^*/\pi.$$

The shape parameters (width/length) vary between 2×10^{-4} and 5×10^{-2}, thus, demonstrating that the calculations used in this study remain well within the cracking range. As far as this study is concerned, very few crack shape parameters in the rock, not subjected to mechanical tests are inferior to 10^{-4}.

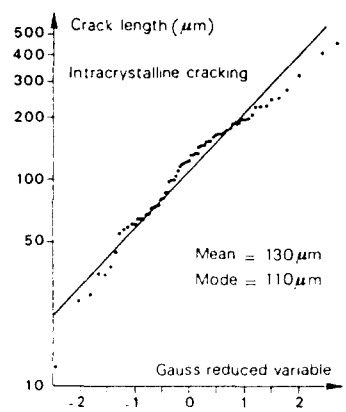


Fig. 1. Intracrystalline cracking distribution of non-treated Remiremont granite.

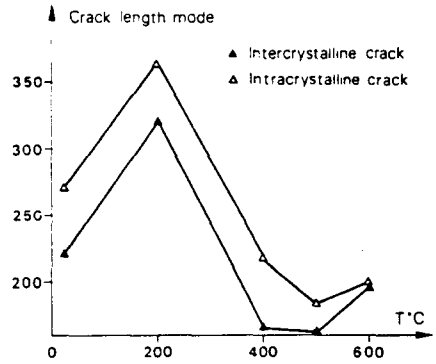


Fig. 2. Intercrystalline and intracrystalline crack length mode of Senones granite as a function of the temperature.

With these data in hand, it is possible to study length, width and shape parameter distributions as well as to calculate the cracking density.

3.2.1. Crack lengths. The length of measured cracks was adjusted according to Galton's laws (Fig. 1). Generally, the application of these laws is relevant for all rocks [8]. The distribution of crack lengths being unimodal, the mode is considered here as the crack's representative value.

With the exception of Senones granite at 200°C, a rather notable similarity of inter- and intracrystalline crack length modes is observed (Fig. 2). Before 600°C, the latter are always more important than the intercrystalline cracks. Up to 600°C, intracrystalline cracking occurs, in particular, within feldspars which propagate in straight lines following the cleavages or the macles. Intercrystalline cracking involves all minerals, quartz being extremely xenomorphic. Therefore, according to the preceding definition, the elementary crack length is very small.

The evolution of Remiremont granite crack length modes is approximately similar in behaviour (Fig. 3). No matter what the intensity of the thermal treatment is, the distribution of crack lengths is almost the same. In fact, crack lengths depend only upon the initial material structure and, in particular, the shape and dimensions of the crystals.

3.2.2. Crack widths. Likewise, by applying Galton's laws [6], crack widths can be adjusted. These widths increase noticeably with pre-cracking temperature. The same is true for intercrystalline (Fig. 4) and intracrystalline cracks. The latter, however, are often less open than the intercrystalline ones. From 200°C upwards, crack widths increase noticeably. However, between 500° and 600°C, such cracks manifest the most significant increase. In the case of Remiremont granite, the openings are more than tripled.

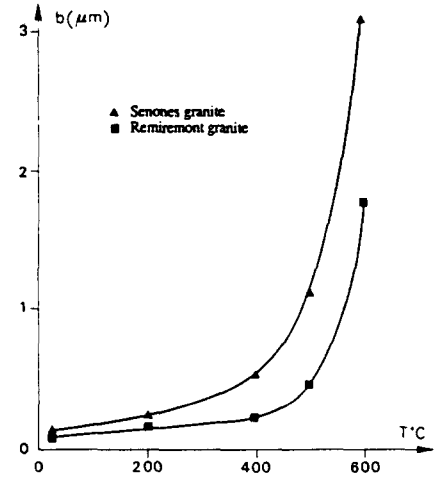


Fig. 4. Intercrystalline crack widths variations of Senones and Remiremont granite as a function of the temperature.

3.2.3. Cracking density. This density is a global parameter which defines a degree of cracking that can be expressed in different ways.

The number of cracks calculated by unit area does not give an accurate measure of the cracking since it favours small cracks. Tapponier and Brace [10] estimated this density in a more reliable way by counting the number of inter- and intracrystalline crack intersections with a network of orthogonal lines. This quantitative stereology technique was developed by Unterwood [11]. For each rock studied, the computer counted the number of crack intersections with a network of regularly spaced orthogonal lines. The resulting data are rather closely related depending upon the two directions. The average is known as *PL* (average of crack intersections by unit length).

In the same way, cracking density can be defined as the ratio of cumulated crack lengths by unit area:

$$D = \frac{\sum a_j}{A},$$

where a_j specifies the crack length j and A the area observed.

The relation of these two definitions for the cracking density of Senones and Remiremont granites is illustrated in Fig. 5. The measure of D and PL densities

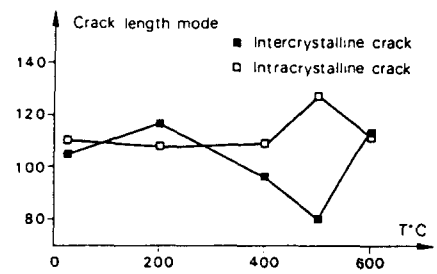


Fig. 3. Intercrystalline and intracrystalline crack length mode of Remiremont granite as a function of the temperature.

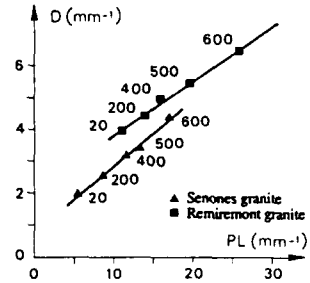


Fig. 5. Correlation between the number of crack intersections *PL* with a counting grid and crack lengths per unit area *D*.

concern the quantity of inter- and intracrystalline crackings. Nevertheless, this correlation is not linear for the entire range of possible values. This, perhaps, is due to the fact that the counting grid is not the same for these two granites.

From now on the crack density value obtained by the summation of crack lengths will be used.

Cracking densities of Senones and of Remiremont non-treated granites are as follows:

	Remiremont granite	Senones granite
Intercrystalline cracking	2.34 mm^{-1}	1.01 mm^{-1}
Intracrystalline cracking	1.60 mm^{-1}	1.03 mm^{-1}
Total cracking	3.94 mm^{-1}	2.04 mm^{-1}

With the intercrystalline cracks, if one takes account of the most developed crystal boundary length in unit area, the rock with the finest grain size is the most microcracked.

The graph in Fig. 6 shows the evolution of intercrystalline, intracrystalline and total crack densities in relation to thermal treatment. The ratio D/D_0 was plotted, D being the cracked density at temperature T, and D_0 that of the non-treated rock. Generally, it is observed that Remiremont granite is more sensitive to thermal cracking than the Senones granite.

As far as the Remiremont granite is concerned, the intercrystalline cracks always predominate over the intracrystalline cracks. This granite has important intercrystalline cracking potentialities. Its intracrystalline crack lengths are limited due to its small grain-size. Intercrystalline cracking increases fairly regularly up to 600°C. By contrast, the intracrystalline cracking increases fairly weakly until 400°C, after which it undergoes a marked acceleration.

This evolution of crack densities is rather similar for the Senones granite. Here, the intercrystalline cracking is stronger than that of intracrystalline up to 500°C. Since Senones granite (with large crystals) intercrystalline cracking potentialities are limited, their growth is quite slight between 500° and 600°C. On the other hand, the intracrystalline cracks increase strongly between 500° and 600°C. Probably, this is a consequence of dilatational differences within the crystals for several reasons (such as mineralogical inclusions, fluid inclusions and zonations, in particular, in the plagioclase). All these factors result in unequal dilatations and induce tensile stresses that provoke the formation of intracrystalline cracks.

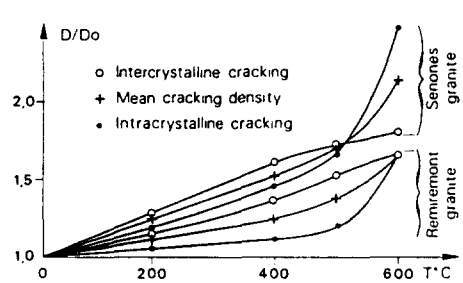


Fig. 6. Cracking density variation as a function of the temperature. D: cracking density at temperature T; D_0: cracking density of the non-treated rock.

For non-treated rock the crack lengths density (global cracking) is equal to:

2.15 Senones granite
1.66 Remiremont granite

It is clear here, that this parameter cannot grow indefinitely. The intercrystalline cracking is limited by the length of the intercrystalline boundaries. When this cracking is entirely open, it results in a material without cohesion. This occurs at 800°C for Senones granite and at 900°C for Remiremont granite.

The intracrystalline cracking potentials become only apparent from 600°C upwards. In addition, to study correctly the thermal microcracking phenomena, it is essential to take into account the crack openings in terms of crack porosity.

3.2.4. Estimation of crack porosity. The calculation of crack porosity from measurements necessitates linking of bidimensional characteristics (length, width) to tridimensional ones (volumes). Crack porosity can be measured from bidimensional measurements if isotropy is assumed. Porosity is thus defined by the sum of area voids per unit area. All other attempts to obtain the tridimensional assessment give deviating results [9].

The calculation of the sum of the area of voids depends upon the model selected to represent the crack's shape. The porosity of a crack model composed of cylindrical tubes may be expressed as:

$$P1 = \frac{\sum a_j b_j}{A},$$

where A is the total area, a_j the length and b_j the diameter of the tube.

In the case of an elliptic crack model, the elementary void resembles a long ellipse with long axis a_j and small axis b_j. In this case, porosity is equal to:

$$P2 = \frac{1}{A}\sum \frac{\pi}{4} a_j b_j = \frac{\pi}{4A}\sum a_j b_j.$$

When dealing with mechanical properties, the authors generally consider the elliptical crack model. In fact, this model is adequate for intracrystalline cracks. The cylindrical tube model, however, seems to be better adapted to the morphology of intercrystalline cracks. The difference between $P1$ and $P2$ porosities is only the factor $\pi/4$, which is small taking into account the unreliability of bidimensional approximation and the measurement errors. SEM measurements of non-treated granites give a crack porosity of 3.2×10^{-4} for Senones granite and 6.7×10^{-4} for Remiremont granite. Hadley's estimation of the crack porosity of Westerley granite [9], namely, 1.3×10^{-3}, is the only available source for comparison.

Increase in porosity induced by cracking is extremely sensitive to thermal treatment. Figure 7 shows the ratio η/η_0, where η specifies the crack porosity at temperature T and η_0 the initial crack porosity. The increase in porosity is higher between 500° and 600°C. These results

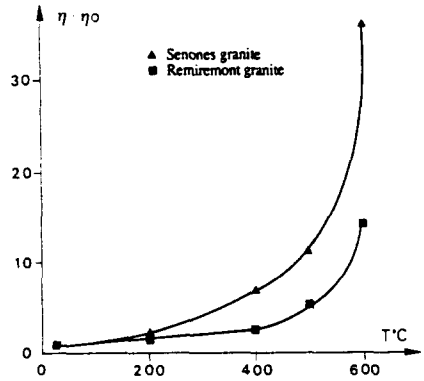


Fig. 7. Crack porosity increase as a function of the temperature. η: crack porosity at temperature T; η_0: crack porosity of the non-treated rock.

will be later compared with those obtained by compressibility tests.

4. STUDY OF CRACKING USING THE COMPRESSIBILITY TEST

Rock strain measurements under hydrostatic pressure show that the relative volume decrease by unit pressure (compressibility) is not always constant. Adams and Williamson, (for example, see Walsh [12]), were the first to attribute this inelastic behaviour to the presence of mismatches and to their closing under pressure. Basically, these mismatches are microcracks and not pores because the pressures required to close the spherical cavities are extremely high.

The compressibility test, thus, enables the crack porosity of a rock to be determined. Furthermore, with the aid of compressibility curves and closure hypotheses [12–14], it is theoretically possible to calculate certain parameters in relation to microcracking. These parameters are described below.

4.1. Crack porosity

Figures 8 and 9 demonstrate the results of tests carried out on Senones and Remiremont granites. Their initial crack porosities η_0 are 0.26×10^{-3} and 0.63×10^{-3} respectively. The behaviour of both granites is similar, in particular, between 500° and 600°C where a significant deviation occurs. Figure 10 illustrates the crack porosity

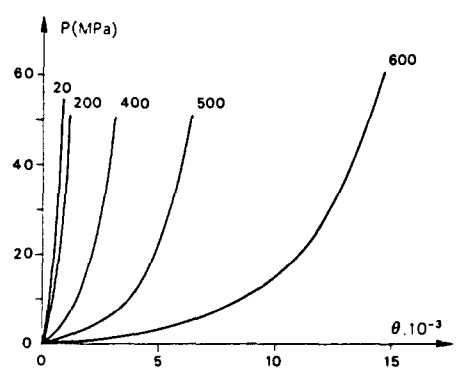


Fig. 8. Pressure–volumetric strain curves of Senones granite as a function of the temperature.

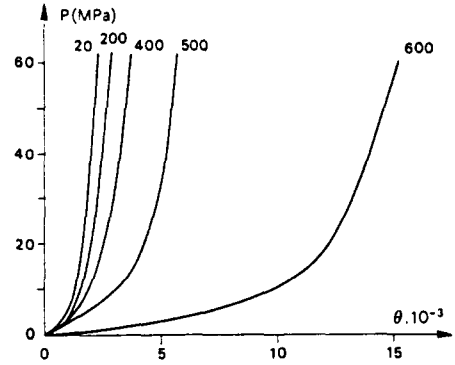


Fig. 9. Pressure–volumetric strain curves of Remiremont granite as a function of the temperature.

evolution η/η_0 in relation to temperature, where η specifies the crack porosity at temperature T and η_0 the initial crack porosity. After 400°C, the increase of crack porosity is very pronounced; thereafter, it accelerates strongly between 500° and 600°C. In the Remiremont granite this cracking evolution with respect to a given temperature is not so marked as in the Senones, the latter rock being less microcracked from the beginning. The results were compared with those of Vosgian sandstone, whose crack porosity remains virtually constant during heating. This is because Vosgian sandstone has a significant pore porosity.

Compressibility tests give crack porosity values that can be compared to those resulting from dimensional measurements carried out using SEM. The methods show particularly good agreement between the values obtained. This once more confirms the viability of using SEM measurements (Figs 7 and 10).

4.2. Interpretation of the compressibility curve

The aim of Walsh's research [12], (further developed by Morlier [13, 14]) was to interpret the $\eta(P)$ curve, showing the porosity variation as a function of pressure. This consequently led to obtaining the microcracking spectrum as a function of parameter α (width/length).

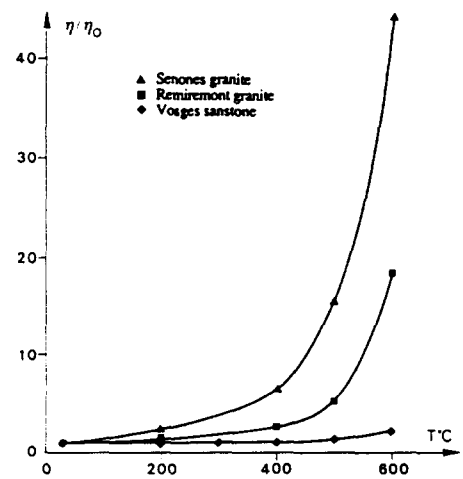


Fig. 10. Crack porosity as a function of the temperature. η: porosity in relation to the rock treated at temperature T; η_0: non-treated rock porosity.

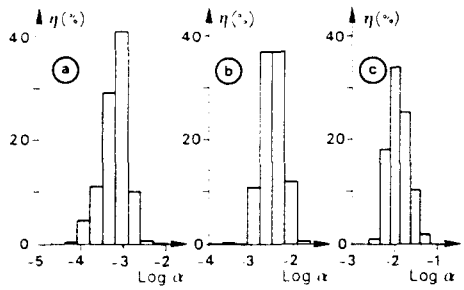


Fig. 11. Histograms of Senones granite porosity (SEM measurements). (a) non-treated rock; (b) rock at 400°C; (c) rock at 600°C.

SEM measurements of granite show significant differences in the order of magnitude of α. By using the compressibility curves according to Morlier's theory [13], it was observed that the distribution mode of porosities is 5×10^{-4} where α_{max} was 3.6×10^{-3} in the Senones granite pre-cracked at 600°C [6]. For the same granite, SEM gave a mode of shape parameters of 1.5×10^{-2} where α_{max} was 6×10^{-2}. The histogram of porosity distribution as a function of the shape parameter was plotted for the Senones granite, the non-treated rock and treated rocks at 400° and 600°C (Fig. 11). Porosity distribution of the non-treated rock is entirely comparable to those obtained from the Westerly granite by Hadley [9] using SEM measurements. The porosity mode distribution of Westerly granite corresponds to the cracks of α between 10^{-3} and 1.5×10^{-3}. However the crack porosity at α between 10^{-4} and 10^{-3} was very small. The histogram of Fig. 11 shows an increase of α as a function of temperature. The influence of crack shapes that are shown here to be non-elliptical can explain, partly, the poor agreement between theoretical and experimental results.

In any event, the compressibility test can only provide indications concerning the morphology of cracks present in a rock. The crack porosity estimation η_0 can, however, be considered useful.

5. MECHANICAL BEHAVIOUR

The study of thermally pre-cracked rocks is of double interest. On the one hand, it reveals principally the modifications that were brought about by rock temperature variations. On the other hand, it provides, after a thermal cycle, samples with basically the same

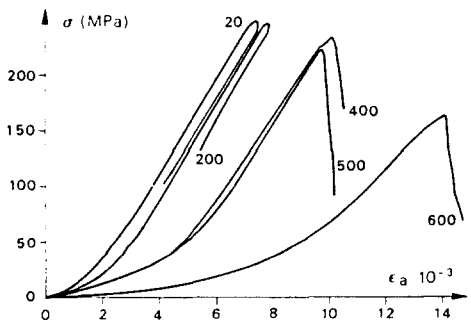


Fig. 12. Remiremont granite stress–axial strain curves at different pre-cracking temperatures; $\dot{\epsilon} = 10^{-5}\,s^{-1}$.

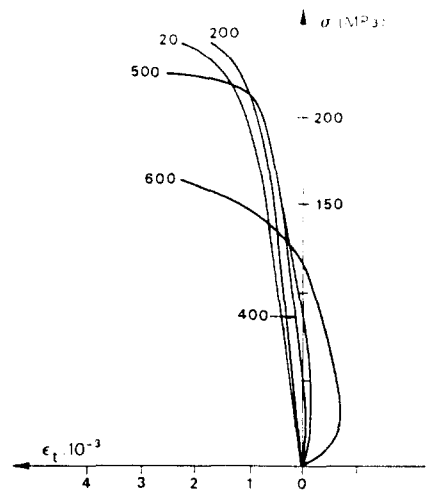


Fig. 13. Remiremont granite stress–lateral strain curves at different pre-cracking temperatures; $\dot{\epsilon} = 10^{-5}\,s^{-1}$.

mineralogy. Their elastic parameters are then solely dependent upon the degree of cracking created. All the experiments on treated samples were carried out when they were cooled down to room temperature.

5.1. *Uniaxial compression test results*

5.1.1. Axial, lateral, volumetric stress–strain curves. The recordings of axial and lateral strains from the strain gauges, lead to the calculation of the sample volume variations during the loading. The stress–lateral strain ϵ_t and the volumetric curves result directly from measurements recorded by 3 axial and 3 lateral gauges (20 mm of active length) mounted at the mid-height of the samples. By preference, in the case of the stress–axial strain curves σ–ϵ_a, the axial strain recordings were carried out with two LVDTs, placed between platens of the press, in order to obtain the post-peak curves, as this is not possible with strain gauges. Out of 3–5 tests carried out for each temperature, one example is given herewith.

Figures 12–14 represent the axial, lateral and volumetric stress–strain curves of Remiremont granite.

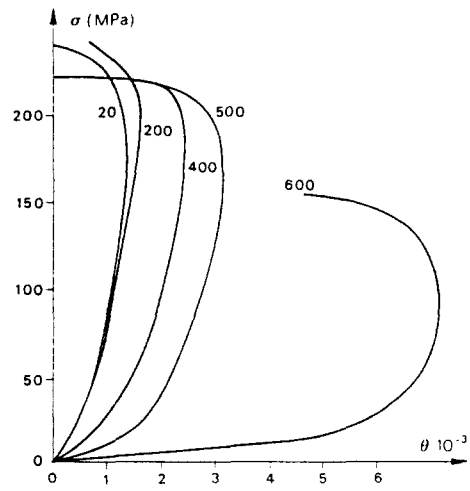


Fig. 14. Remiremont granite stress–volumetric strain curves at different pre-cracking temperatures; $\dot{\epsilon} = 10^{-5}\,s^{-1}$.

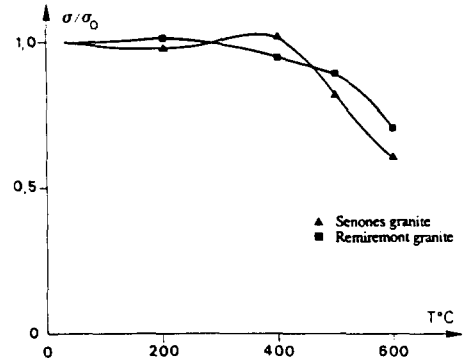


Fig. 15. Compressive strength variations σ as a function of the thermal cycle's maximum temperature; σ_0: non-treated rock strength.

Curves $\sigma-\epsilon_a$ show the important overall influence of cracking intensity on the ultimate strength and deformability of the material (Fig. 12). With $\dot{\epsilon} = 10^{-5}s^{-1}$, Remiremont non-treated granite shows a Class II behaviour; whereas, with temperature from 400°C upwards, a transition to Class I behaviour is observed. The post-peak curves of samples treated at those temperatures, have a very steep slope. Figure 13 illustrates the stress–lateral strain curves, $\sigma-\epsilon_t$. These curves differ significantly from the elastic behaviour. The lateral strains of treated granites at 200°C are, generally, less than those of non-treated rock. However the overall shape of the curves remain mostly the same. By contrast, at temperatures from 400°C upwards, and at the beginning of loading, negative lateral strains are observed. Curves $\sigma-\epsilon_t$ are generally highly sensitive to cracking. At higher level of heat treatment, the deviation of $\sigma-\epsilon_t$ curves from linearity becomes more pronounced. The same trend is observed for $\sigma-\theta$ curves (Fig. 14).

5.1.2. The variation of mechanical properties. The variations in uniaxial compressive strengths are weak until a temperature of 400°C is attained (Fig. 15). Statistical tests upon the results from 12 samples showed that these variations in uniaxial compressive strength are not significant. From 400°C upwards, however, a significant drop in strength occurs.

Except for Remiremont granite at 200°C, the elastic moduli diminish weakly until 400°C (Fig. 16). The change in moduli is more pronounced at higher temperatures. In the case of Senones granite, the modulus decreases more rapidly. It has already been shown that

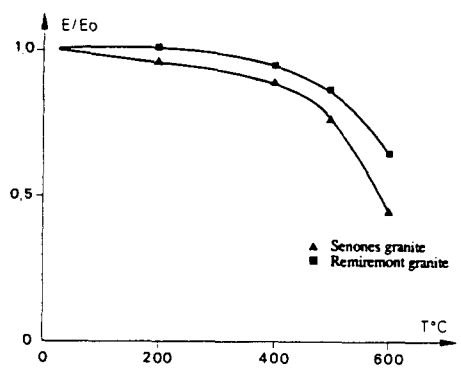


Fig. 16. Elastic modulus E as a function of the thermal cycle's maximum temperature; E_0: non-treated rock modulus.

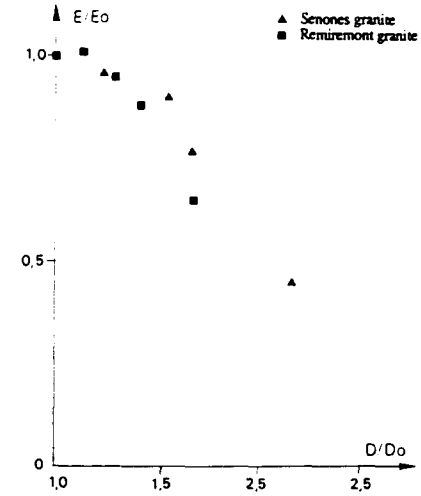


Fig. 17. E/E_0 ratio as a function of crack density; E_0 and D_0 are related to the non-treated rocks.

the latter granite has a more significant crack density and porosity increase as a function of temperature than Remiremont granite. The link between the drop of E/E_0 moduli and the increase of D/D_0 cracking density is relatively satisfactory (Fig. 17).

5.2. The behaviour under tensile loading

The preceding experimental results show that for granites pre-cracked up to a 400°C temperature the compressive strength is a parameter rather insensitive to the increase of cracking. The advantage of the direct tension test is that the cracks are more directly loaded. It is useful to study tensile stress–strain curves in order to compare them to compressive stress–strain curves. Theoretically, for isotropic and homogeneous materials, under compression or tension, Young's moduli and Poisson's ratios are similar.

Direct and indirect tension (Brazilian) tests were carried out. The latter lends itself to criticism because the fracture plane is imposed on the material. Nonetheless, this test is rapid to carry out and results not very dispersed. This test is also influenced by the density of cracking.

5.2.1. Brazilian tests. Figure 18 gives the results of these tests according to the relation of the tensile

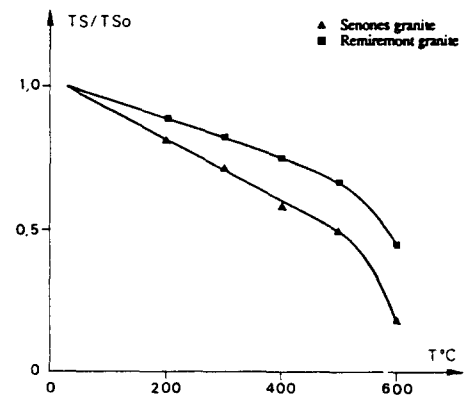


Fig. 18. Tensile strength variation TS as a function of the pre-cracking temperature; TS_0: tensile strength of the non-treated rock (Brazilian test).

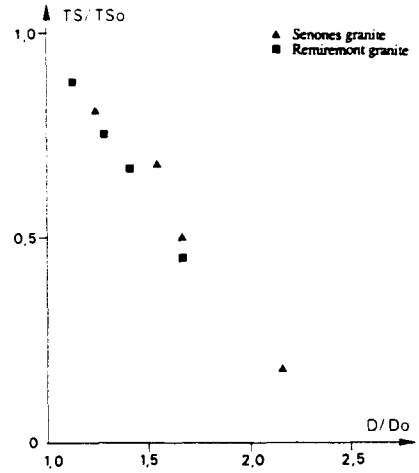


Fig. 19. Relation between tensile strength decreasing and crack density increasing (Brazilian test).

strength after thermal treatment (TS) to the non-treated rock tensile strength (TS_0). The average tensile strength values for non-treated Senones granite and Remiremont granite are 8 and 10 MPa respectively. The tensile strength of Senones granite decreases more rapidly than that of Remiremont granite. The length of cracks formed are more important in the former granite. The tensile strength variation is similar to that found in Westerly and Charcoal granites [2]. The decrease in tensile strength is closely linked to the increase in the density of microcracking (Fig. 19).

5.2.2. Direct tension tests. Figure 20 is an example of the type of axial stress–axial strain curves occurring in Remiremont granite (measured with strain gauges) that result from direct tension tests. The modulus of elasticity (slope of the curve at the origin) is smaller than that of the same rock in uniaxial compression. The stress–lateral strain curve of non-treated rock (Fig. 21) determines the Poisson's ratio of 0.12 which is much below that of the rock in compression. All stress–lateral strain curves for pre-heated rocks have a lateral dilatation. At the beginning of loading, samples previously subjected to temperatures of 200° and 400°C demonstrate negative lateral strains and positive ones after a certain stress level. The stress level at which the transition occurs is much lower than that when the material is more cracked. The sample dilatates entirely from 600°C. Senones granite behaves in a virtually identical manner [6].

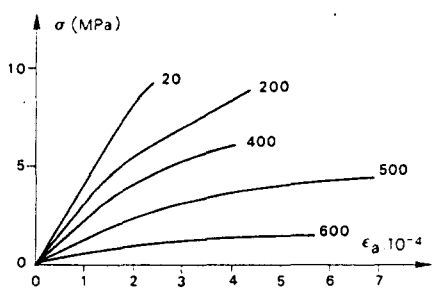


Fig. 20. Remiremont granite tensile stress–axial strain curves at different pre-cracking temperatures (direct tension test).

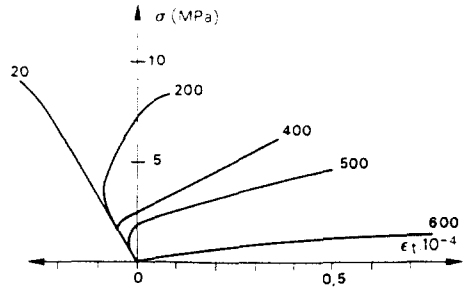


Fig. 21. Remiremont granite tensile stress–lateral strain curves at different pre-cracking temperatures (direct tension test).

6. DISCUSSION

The particular phenomenon occurring with stress–lateral strain curves of thermally cracked rocks is that the sample dilatates laterally, at a certain stress level in tension (Fig. 21). At the beginning of compression loading the sample contracts axially as much as laterally (Fig. 13). Since these behaviour elastic deviations are proportional to the maximum temperature reached during the thermal cycle it seems evident to attribute its genesis to thermal cracking. Observations show that this cracking has the character of being dominantly intercrystalline.

Such lateral strain phenomena are of a opposite sign to the normal and give negative Poisson ratios. The mathematical theory of elasticity (for isotropic materials) imposes on the modulus of Young E, and the Poisson ratio v the following conditions:

$$E > 0; \quad -1 < v < 0.5.$$

Consequently, negative values of v are theoretically possible. However, for real isotropic materials, the experiment always results in positive values for the Poisson ratio.

The two explanations possible for these abnormal signs of lateral strain were verified to be neither an effect of surface, nor of the strain gauges. A 35 mm diameter sample, cores in a $10 \times 10 \times 10$ cm cube of granite, previously subjected to thermal treating at 600°C, showed lateral strains of an abnormal sign. To verify the measurements obtained by strain gauges, two horizontal LVDTs, were used against the sample at its mid-height.

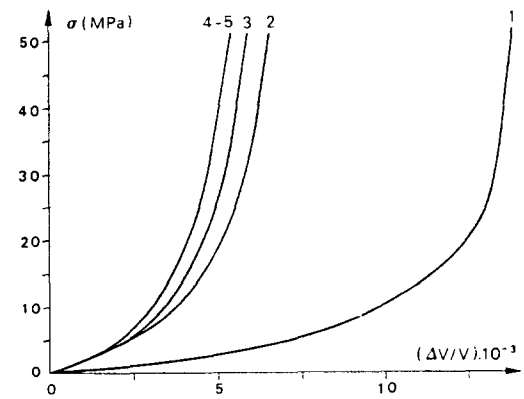


Fig. 22. Pressure–volumetric curves of 600°-treated Remiremont granites carried out on the same sample.

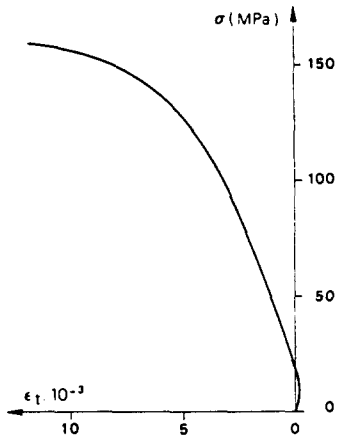


Fig. 23. Remiremont granite stress–lateral strain after partial closure of cracks.

If it is accepted that thermal cracking provokes abnormal lateral strains, it must be shown, by closing these cracks, that a normal behaviour of σ–ϵ_t is obtained. In this study, the compressibility test was used, as it is possible to obtain different crack porosities for each cycle of loading. A Remiremont granite sample, previously pre-cracked at 600°C and equipped with strain gauges was subjected to a maximum pressure of 50 MPa. Volume variation curves (Fig. 22) demonstrate that the closure of all cracks is possible and that the pressure volume curve does not change from the fourth cycle. For the first cycle, porosity of cracks is 1.25×10^{-2} and from the fourth cycle 0.4×10^{-2}. Therefore, a relatively important structural rearrangement occurs, without crack closures being complete. A further uniaxial compression test, carried out on the same sample gave a stress–lateral strain curve (Fig. 23) very different from that obtained for this granite pre-cracked at 600°C. The negative lateral strains are greatly reduced. Part of the curve σ–ϵ_t is linear. The closure of a significant proportion of cracks, created by thermal treatment results in an almost normal behaviour for σ–ϵ_t. When there is an intercrystalline cracking, the minerals are no longer interdependent on their neighbours. According to certain tension loads, they are able to move. A crystal's rotational (Fig. 24) and translational movements can induce transversal displacements and thereby, positive transversal strains [15]. Such structural rearrangements occur with displacements and when sufficient stress is applied. Structural rearrangements, in compression, occur at the beginning of loading since the first stage of the test is known as the closure of cracks phase. In a slightly cracked rock, more or less entire closures take place for cracks perpendicular to the stress direction and cannot result in negative lateral strains. Whereas, it is easy to find examples of structural rearrangements giving positive strains. In compression it is more difficult, it not impossible, to find rearrangements which correspond to negative strains.

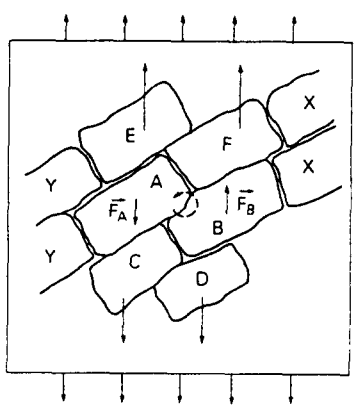


Fig. 24. Rotational movements of grains.

This gives rise to the supposition of the existence of residual stresses in the rock induced during recooling process. All grain boundaries are not broken, since even at 600°C a cohesion remains (and a tension strength). When the temperature is decreased considerably, dilated minerals will contract. Since certain crystals still remain linked, such a contraction does not occur freely. It is maintained, here, that the return towards original dimensions and shapes can be upset, inducing residual compressive stresses in the rock. Eventually, at the moment of loading, these will introduce negative lateral strains. Subsequent cycles will free these stresses while at the same time closing a part of the cracks. The curve σ–ϵ_t in compression will, then, have a normal aspect.

7. CONCLUSIONS

By characterizing the microcracks through SEM and compressibility tests it is possible to draw the following conclusions:

(a) SEM measurements of crack lengths and widths result in the determination of a parameter of cracking density and a calculation of a crack porosity.
(b) SEM observations and measurements show that crack length does not vary in the granites. The intracrystalline cracks are always longer than the intercrystalline ones.
(c) The shape of cracks is not consistent with an elliptical model, rather they may be approximated well with a network of tubes, having rough surfaces. Only, the crack tips are sharp.
(d) Crack widths increase with temperature.
(e) The comparison between crack porosity, calculated by morphological observations carried out with SEM, and that determined by the compressibility test, yields reasonably similar results. Whereas this test is reliable for the characterization of the volume of cracks, their dimensional characteristics cannot be assessed by it.

The study of the mechanical behaviour of thermally cracked granites show that cracking density increases with the intensity of thermal treatment. The Young's moduli in compression and tension and corresponding ultimate strengths are reduced accordingly, as a function of the maximum temperature of each cycle.

Likewise, this study demonstrates that elastic modulus is more sensitive to cracking increase than is compressive strength. A fairly good relation exists in the granites between the modulus decrease and the cracking density increase. By contrast, compression strength is more dispersed, requiring a high quantity of cracks to appreciably decrease it. Note must be taken that thermal cracks are small and essentially intercrystalline. The maximum mismatch length most unfavourably orientated in a rock composed of crystals is of primordial importance. Frequently, this may be a pre-existing mismatch such as a large crack in a feldspar. With respect to compressive strength, it is not the degree of homogeneous cracking but of heterogeneity that is the significant factor.

Lateral strains of abnormal sign appear. These are due to the development of intercrystalline microcracking and increase with the thermal cycle intensity. It is highly probable that these abnormal strains are induced by residual stresses created during re-cooling. However, abnormal lateral strains in tension are explained by the structural rearrangments caused by the existence of intercrystalline cracking.

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